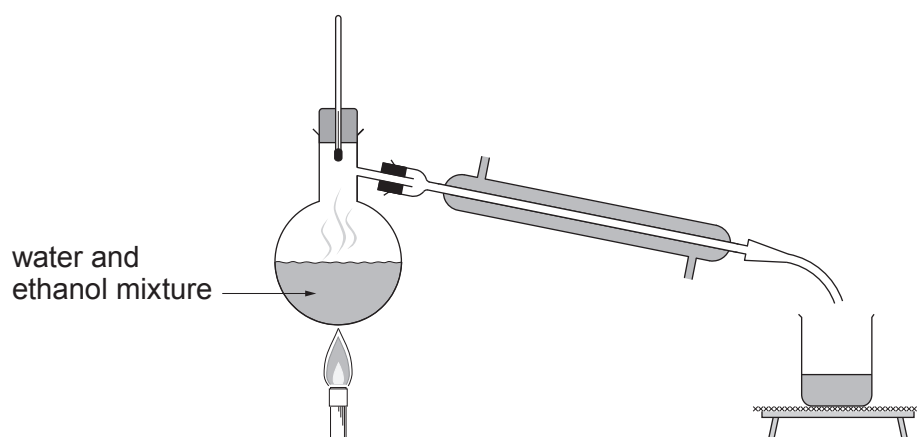


2. (a) A teacher carried out an experiment to show how to separate a mixture of ethanol and water.



- (i) Choose from the box the name of the method used.

[1]

chromatography

filtration

distillation

evaporation

Method

- (ii) The following sentences show the steps involved in this method of separation.

- A Ethanol vapour cools and condenses
- B Ethanol boils and evaporates
- C The mixture of water and ethanol is heated
- D Ethanol is collected

Put the steps in the correct order. Use the letters **A-D**.

[1]

.....

- (iii) Tick (✓) the box below that explains why it is the water that is left in the flask.

[1]

the boiling point of ethanol is equal to the boiling point of water

☐

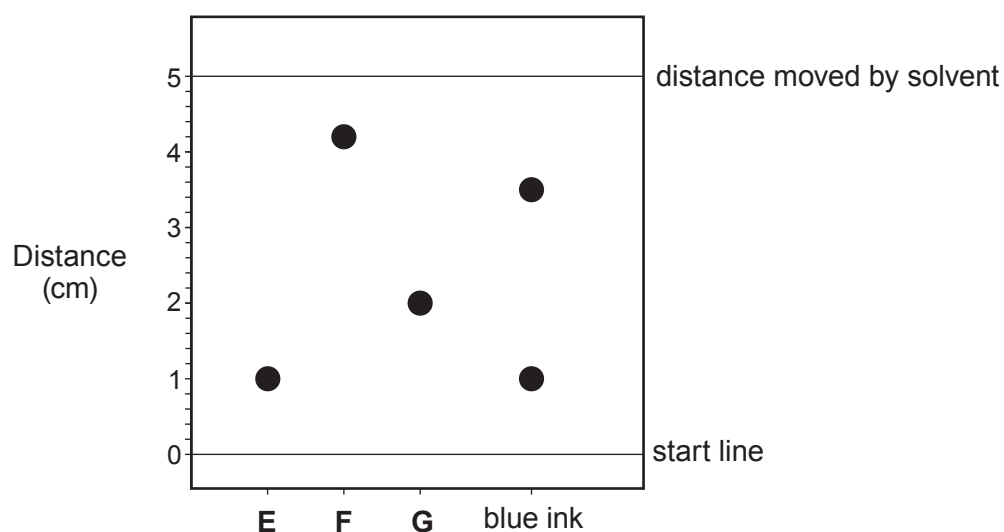
the boiling point of ethanol is lower than the boiling point of water

☐

the boiling point of ethanol is higher than the boiling point of water

☐

- (b) A student carried out an investigation to find out whether pigments **E**, **F** and **G** are present in a blue felt pen. The results are shown below.



- (i) State what the results tell you about the blue ink. [2]

.....

.....

- (ii) Calculate the R_f value of pigment **F** using the equation given. [2]

$$R_f = \frac{\text{distance moved by pigment}}{\text{distance moved by solvent}}$$

$$R_f = \dots\dots\dots$$

